

LAB-1: Mining and Profiling a Software Repository

Objective:

By the end of this lab, you should be able to:

- Understand a software repository as a source of structured and unstructured software engineering data.
- Clone/access an open-source GitHub repository programmatically.
- Traverse repository files and identify source-code files.
- Extract basic repository-level and file-level metrics.
- Use Git history to extract basic development information.
- Produce a machine-readable dataset that can be used in subsequent labs.
- Recognize the difference between source code analysis and repository mining.

Lab Tasks:

1. Repository Inventory- Clone the repository and determine
 - Repository name
 - Number of files
 - Number of source-code files
 - Programming languages
 - Number of directories
 - Total LOC
 - LOC per source file
 - Largest source files
 - File-type distribution
2. File level Dataset
 - Create a csv file like below
 - file_path,language,extension,loc,size_bytes
 - src/main/User.java,Java,.java,184,6212
 - src/main/Login.java,Java,.java,237,8192
 - src/utils/Parser.java,Java,.java,421,15421
3. Mine Git History and make repository level observation around following
 - Number of commits
 - Number of contributors and most active contributor
 - Files changed most frequently
 - Commits per month
 - Average files changed per month
 - Average file addition or deletion

Output:

1. Prepare the statistics into json file and commit
2. Prepare csv files as datasets and commit
3. Update the README.md file and commit

Repositories:

1. <https://github.com/pallets/flask>
2. <https://github.com/psf/requests>
3. <https://github.com/pytest-dev/pytest>
4. <https://github.com/fastapi/fastapi>
5. <https://github.com/scikit-learn/scikit-learn>